2 Mapping physical systems to a digital representation

This chapter covers

- Deciding what aspects of the physical world to capture digitally
- Choosing information sources to build a digital model
- Extracting and digitizing information from these information sources
- Relating objects to each other spatially

Building a digital twin begins with the fundamental questions of what aspects of your physical world you should represent digitally, and how accurately you need to represent them. This decision shapes everything that follows, from the sensors you deploy and the data you collect, to the models you build and the insights you can extract. The challenge lies in translating the rich, multi-dimensional complexity of physical systems into digital representations that are both technically feasible and provide business value. Getting this translation right is important because it determines whether your digital twin will be a powerful tool for optimization and prediction, or an expensive collection of data that fails to deliver meaningful results.

This chapter addresses the first step in any digital twin implementation—understanding what to digitize and how you can map the physical world to a digital representation. The core decisions covered here establish the basis for the technical implementations that enable effective data collection and system representation. Without a clear understanding of what you're trying to represent digitally, even the most sophisticated digital representation of your physical system will fail to create an effective digital twin.

Before diving into the technical approaches for digital representation, you must first establish clear objectives that will guide every subsequent decision about what to digitize, how to represent it, and which details matter most for your specific use case.